

Propositional Logic

Objectives

- The lecture will enable you to
- Represent simple facts in the language of propositional logic
- Interpret a propositional logic statement
- Compute the meaning of a compound proposition

Knowledge and intelligence

- Does knowledge has any role in demonstrating intelligent behavior?

How can we represent knowledge in a machine?

- We need a language to represent domain knowledge
- There must be a method to use this knowledge
- Inference mechanism
- Syntax and semantics of a language
 - Laughs (Karim)==??
 - Likes (Farzana, Tania) ==??

Logic is a Formal Language

- Propositional Logic
- Rahima is intelligent - proposition
- Rahima is hardworking – proposition
- If Rahima is intelligent and Rahima is hardworking Then Rahima scores high marks - proposition
-

Elements of PL

- Rahima is intelliegent
- Rahima is hardworking
- Objects and Relations or functions

Towards the syntax

- Intelligent (Rahima) == Rahima is intelligent
- Hardworking (Rahima) == Rahima is hardworking
- p

Towards more syntax

- Let P stand for intelligent (Rahima)
- Let Q stand for Hardworking (Rahima)
- What does $P \wedge Q$ (P and Q) mean ?
- What does $P \vee Q$ (P and Q) mean ?
- $P \wedge Q$, $P \vee Q$ are compound propositions

Syntactic Elements of PL

- Vocabulary
 - A set of propositional symbols (P, Q, R etc.)
 - each of which can be True or False
 - Set of logical operators : $\wedge$ $\vee$. Not, implies
 - often parenthesis () is used for grouping
 - There are two special symbols
 - TRUE (T) and FALSE (F) – these are logical constants

How to form propositional sentence?

- Each symbol (a proposition or a constant) is a sentence
- If P is a sentence and Q is a sentence
- Then
 - (p) is a sentence
 - $P \wedge Q$ is a sentence
 - $P \vee Q$ is a sentence
 - $\neg P$ is a sentence
 - $P \implies Q$ is a sentence
 - Nothing else is a sentence
- Sentences are called world form formulae (wff)

Example of WFFs

- P
- True
- P and Q
- $(P \text{ or } Q) \text{ implies } R$
- $(P \text{ and } Q) \text{ or } R \text{ implies } S$
- Not $(P \text{ or } Q)$
- Not $(P \text{ or } Q) \text{ implies } R \text{ and } S$

Implication

- P implies Q
- If P is true then Q is true
- If it rains then the roads are wet
- Sufficient condition
- If the roads are wet then it rains ???
-

Equivalence

- P bidirectional Q
- Triangle

What does a wff mean -Semantics

- Interpretation in a world
- When we interpret a sentence in a world we assign meaning to it and it evaluates to either True or False

Semantic of compound sentence

- P: Likes (Rahima, Sanjida)
- Q: Knows (Tania, Farzana)
- World: Rahima and Sanjida are friends and Tania and Farzana known to each other
- $P = T, Q = T$
- $P \text{ and } Q = T$
- $P \text{ and Not } Q = F$

Validity of a sentence

- If a propositional sentence is true under all possible interpretation, it is VALID
- Tautology
- Satisfiability
- Inconsistent

Quiz

- Express the following English sentence in the language of PL
 - It rains in July
 - The book is not costly
 - If it rains today and one does not carry umbrella he will be drenched

Quiz

- If P is true and Q is true then are the followings true or false
 - $P \rightarrow Q$
 - $(\neg P \vee Q) \rightarrow Q$
 - $(\neg P \vee Q) \rightarrow P$
 - $P \vee \neg P$

Inference in PL

Objectives

- Infer the truth value of a proposition
- Reason towards new facts, given a set of propositions
- Prove a proposition given a set of propositional facts

Procedure to derive Truth Value

De Morgan's Theorem

Earlier quiz

- Solution

Reasoning

- P: It is the month of July
- Q: It rains
- R: $P \rightarrow Q$
- If it is the month of July then it rains
- It is the month of July
- Conclude: It rains

Reasoning –Modus Ponens

- Modus Ponens is very much a valid inference rule prove

Valid inference rule

- Thus irrespective of meaning Modus ponens allows us to infer the truth of Q
- Modus ponens is an inference rule that allows us to deduce the truth of a consequent depending on the truth of the antecedent

Inference rules can be mechanically
applied

Satisfiability

- Remember: An interpretation is a mapping to a world
- A sentence is SATISFIABLE by an interpretation
- If under that interpretation the sentence evaluate to true

Satisfiability

- If no interpretation makes all the sentences in the set to be true
- Then the set of sentences is UNSATISFIABLE or INCONSISTENT

Entailment

- This concept is necessary to understand the logical entailment
- We often say this is a logical consequence of that
- We logically follow that
- What does it mean by logically follow?
- This is essentially captured in the idea of entailment
- If something entails something logically we say entailment

Entailment

- If a sentence S_1 has a value True for all interpretation that make all sentences then
- $S \models S_1$
- S_1 logically follows from S
- S_1 is a logical consequence of S
- S logically entails S_1

- If new proposition logically follows from set of sentences then we can conclude new proposition should be true
- We have seen Modus Ponens is a very special form of inference mechanism, we shall now introduce a very important inference mechanism which is known as RESOLUTION
- Before that we need to know

Clauses: A Special Form

- Literal: A single proposition or its negation
- A clause is a disjunction of literals

Converting a compound proposition to the clausal form

- Disjunctive normal form (DNF)
- Step 1: eliminate implication sign
- Step 2: eliminate double negation and reduce scope of “not” signs (De Morgan law)
- Step 3: convert to conjunctive normal form by using distributive and associative law
- Step 4: Get the set of clauses

Resolution – A technique of inference

- A sound inference mechanism
- Principle
- Suppose x is a literal and $s1$ and $s2$ are two sets of propositional sentences represented in clausal form
- If we have $(x \vee s1)$ AND $(\neg x \vee s2)$
- Then we get $s1 \vee s2$
- Here $s1 \vee s2$ is the resolvent
- x is resolved upon

An example

- If a triangle is equilateral then it is isosceles
- If a triangle is isosceles then two sides AB and AC are equal
- If AB and AC are equal then angle B and C are equal
- ABC is an equilateral triangle
- Angle B is equal to angle C - prove

Convert the sentences into PL

Convert the sentences into clausal
form

Proof by refutation

- To prove
- Angle B is equal to Angle C
- Let us disprove
- not equal
- Let us try to disprove this

Procedure for Resolution

- Convert given proposition into clausal form
- Convert the negation of the sentence to be proved into clausal form
- Combine the clauses into a set
- Iteratively apply resolution to the set and add the resolvent to the set
- Continue until no further resolvents can be obtained or a null clause is obtained

Some problems of PL

quiz

- Consider the following sentences
- 1. mammals drink milk
- 2. Man is mortal
- 3. Man is mammal
- 4. Tom is a man
- 5. Prove Tom drinks milk
- 6. Prove Tom is mortal
- Represent all the sentences in the clausal form
- prove (5) and (6) using modus ponens
- Prove (5) and (6) using resolution

First Order Logic

Objective

- Formulate more types of sentences in logic
- Write correct predicate logic formulae

Limitation of PL

- Consider the following argument
 - All dogs are faithful
 - Tommy is a dog
 - Therefore, Tommy is faithful?????
- How to represent and infer this in PL?

Limitation of PL

- Karim is a hardworking student
 - Hardworking (Karim)
- Karim is an intelligent student
 - Intelligent (Karim)
- If Karim is hardworking and Karim is intelligent
Then Karim scores high marks
- $\text{Hardworking (Karim)} \wedge \text{Intelligent (Karim)} \rightarrow \text{Scores_High_Marks (Karim)}$

Limitation of PL

- What about Rahim and Salim?
- If we could write instead
- All students who are hardworking and intelligent scores high marks???
- For all x such that x is a student and x is intelligent and x is hardworking then x scores high marks
 - x is a variable

The problem of infinite model

- In general, propositional logic can deal with only a finite number of propositions
- If there are only three dogs Tommy, Jimmy and Laika, then
 - T: Tommy is faithful
 - J: Jimmy is faithful
 - L: Laika is faithful
 - All dogs are faithful $T \wedge J \wedge L \dots\dots$
- What if there is an infinite number of dogs?

FOL/Predicate Logic

- It is the generalization of PL that allows us to express and infer arguments in infinite models like
 - All men are mortal
 - Some birds cannot fly
 - At least one planet has life on it

What added in FOL?

- Variable, quantifier
- Syntax of FOL Can be determined in terms of
 - Terms
 - Predicate
 - quantifier

Terms

- A term denotes some object other than true or false
 - Tommy is a dog
 - All men are mortal
 - Tommy and men are terms
- Constant
- Terms in function

Terms: Constant & Variable

- A constant of type W is a name that denotes a particular object in a set W
 - Example: 6, Tommy etc
- A variable of type W is a name that can denote any element in the set W
 - Example: $n \in \mathbb{N}$ denotes a natural number
 - D denotes the name of a dog

Terms: Function

- A functional term of arity n takes n objects of type W_1 to W_n as inputs and returns an object of type W
 - $f(w_1, w_2, \dots, w_n)$
 - $\text{Plus}(3, 4) = 7$
 - Plus is a functional term and $3, 4, 7$ are constant terms

Functions: Example

- Let plus be a function that takes two arguments of type Natural Number and returns a Natural Number
- Valid functional terms:
 - Plus (2,3) plus (5, plus (7,3))
 - Plus (plus(100,plus(1,6)), plus (3,3))
- Invalid functional terms:
 - Plus (0, -1) plus (1.2,3.1)

Functions with variable arguments

- `Plus(x,y)`
- `Prod (x,y,z)`

Predicates

- Predicates are like functions except their return type is true or false
- Example:
 - **gt**(x,y) is true iff $x > y$
 - Here **gt** is a predicate symbol that takes two arguments of type natural number
 - **gt**(3,4) is a valid predicate but **gt**(3,-4) is not

Types of predicates

- A predicate with no variable is a proposition
 - Tommy is a dog
- A predicate with one variable is called a property
 - $\text{dog}(x)$ is true iff x is a dog
 - $\text{mortal}(y)$ is true iff y is mortal

Formulation of predicates

- Let $P(x,y,\dots)$ and $Q(x,y,\dots)$ are two predicates
- Just remember how we have talked about wff
- Then so are
 - $P \wedge Q$
 - $P \vee Q$
 - $\neg P$
 - $P \rightarrow Q$
 - Here P and Q are two predicates

Predicate Examples

- If x is a man then x is mortal
 - $\text{man}(x) \rightarrow \text{mortal}(x)$
 - $\neg \text{man}(x) \vee \text{mortal}(x)$ [clausal form]
 - The above are not propositions because they contain variable x
- If n is a natural number then n is either even or odd
 - $\text{Natural}(n) \rightarrow \text{even}(n) \vee \text{odd}(n)$

Quantifiers

- There are two basic quantifiers in FOL
 - $\forall$ “for all” universal quantifier
 - $\exists$ “there exists” Existential quantifier

Universal quantifiers

- All dogs are faithful
 - Faithful (x): x is faithful
 - Dog (x): x is a dog
 - $\forall x (\text{dog}(x) \rightarrow \text{faithful}(x))$

Universal quantifiers

- All birds cannot fly
 - Fly (x): x can fly
 - Bird (x): x is a bird
 - $\neg(\forall x (\text{bird}(x) \rightarrow \text{fly}(x)))$

Quiz problem revisited

- Mammals drink milk
- Man is mortal
- Man is mammal
- Tom is a man

Existential quantifier

- At least one planet has life on it
 - Planet (x): x is a planet
 - haslife (x): x has life on it
 - $\exists x (\text{planet}(x) \wedge \text{haslife}(x))$

Duality of quantifiers

- All men are mortal $\Leftrightarrow$ No man is immortal
- There exist birds that can fly $\Leftrightarrow$ it is not the case that all birds cannot fly

Sentences

- A predicate is a sentence
- If sen , sen' are sentences x a variable, then
 - (sen) , $\neg \text{sen}$, $\exists x \text{ sen}$, $\forall x \text{ sen}$, $\text{sen} \wedge \text{sen}'$, $\text{sen} \vee \text{sen}'$,
 $\text{sen} \rightarrow \text{sen}'$ are sentences
- Nothing else is a sentence
- Compare with PL

conclusion

- FOL is an extension of PL
- More powerful than PL with the inclusion of variables and quantifiers, resulting capturing of sentences in much more natural way which is not possible in PL
- We can do it in PL but we need some kinds of artificial maneuvering but without that we can do that FOL
- Show the similarity of PL and FOL and the extra power of FOL

Exercise

- Some dogs bark
- All dogs have four lags
- All barking dogs are irritation
- No dogs purr
- Fathers are male parents with children
- Students are people who are enrolled in courses

Reasoning in FOL

Example of sentences

- $\text{Birthday}(x,y) \rightarrow x \text{ celebrates birthday on date } y$
- $\forall y \exists x \text{ Birthday}(x,y)$
- For all dates, there exists a person who celebrates his/her Birthday on that date. This is – “Everday someone celebrates his/her Birthday”

Examples of sentences

- Brother (x,y) — x is y 's brother

- Loves (x,y) — x loves y

— $\forall x \forall y \text{ Brother } (x,y) \rightarrow \text{loves } (x,y)$

Everyone loves (all of) his/her brothers.

Let $m(x)$ represent mother of x then

- “everyone loves his/her mother” is
— $\forall x \text{ Loves } (x, m(x))$ [x is related to everyone]

Examples of sentences

- Any number is the successor of its predecessor

$\text{Succ}(x), \text{pred}(x)$

$\text{equal}(x, y)$

$\forall x \text{ equal}(x, \text{Succ}(\text{pred}(x)))$

Alternative representation

- The previous example can be represented succinctly as
- $\forall x (\text{succ}(\text{pred}(x)) = x)$
- Equality operator is not allowed in predicates
- Therefore the above sentence is not valid

FOL with Equality [extended]

- In FOL with equality, we are allowed to use the equality sign ($=$) between two functions
- This is just for representational ease
- We modify the definition of sentence to include equality as
term = term is also a sentence

Exercise revisited

- Some dogs bark
 - $\exists x (\text{dog}(x) \wedge \text{bark}(x))$
- All dogs have four legs
 - $\forall x (\text{dog}(x) \rightarrow \text{have_four_legs}(x))$ [scope of x?/?/role of parenthesis/unary predicate]
 - $\forall x (\text{dog}(x) \rightarrow \text{legs}(x,4))$ [legs is a binary predicate]
- All barking dogs are irritating
 - Do yourself

Exercise revisited conti...

- No dogs purr
 - $\neg \exists x (\text{dog}(x) \wedge \text{purr}(x))$
 - Would there be any alternative representation???
 - Think???
 - think duality???
 - any existential quantifier can also be converted into universal quantifier
- Fathers are male parents with children
 - $\forall x (\text{father}(x) \rightarrow \text{male}(x) \wedge \text{has_children}(x))$
- Students are people who are enrolled in courses
- Do yourself

Inference Rules

- Universal Elimination

$\forall x$ Likes (x, flower)

substituting x by Karim gives

likes (Karim, flower)

The substitution should be done by a constant term

Inference Rules

- Existential Elimination (Skolemization)

$\exists x \text{ Likes } (x, \text{flower}) \rightarrow \text{likes } (\text{Person}, \text{flower})$

As long as person is not in the knowledge base.

- Existential Introduction

$\text{Likes } (\text{Anis}, \text{flower})$

can be written as

$\exists x \text{ Likes } (x, \text{flower})$

Reasoning in FOL

- Consider the following problem:
- If a perfect square is divisible by a prime P , then it is also divisible by square of P .
- Every perfect square is divisible by some prime.
- 36 is a perfect square
- Does there exist a prime Q such that square of Q divides 36?

Representation in FOL

1. If a perfect square is divisible by a prime P , then it is also divisible by square of P .
2. Every perfect square is divisible by some prime.
3. 36 is a perfect square
4. Does there exist a prime Q such that square of Q divides 36?

Inferencing

- From 2 and universal elimination
- 4
- From 4 and existential elimination
- 5
- From (1) and (5)
- 6
- From 5 and 6
- 7
- From 7 and existential introduction

Inferencing (contd.)

- However, inferencing would have been easier if the knowledge base consists of the following sentences
 - Perfect_sq (36)
 - Prime (p)
 - Divides (36,p)
 - $\forall x,y$ perfect_sq (x) $\wedge$ prime (y) $\wedge$ divides (x,y) $\rightarrow$ divides (x, square (y))

Horn sentences

- Atomic sentences/clauses is a horn sentence
 - $\text{Perfect_sq}(36)$
- Implication with a conjunction of atomic sentences on the left and a single atom on the right
 - $\forall x, y \text{ perfect_sq}(x) \wedge \text{prime}(y) \wedge \text{divides}(x, y) \rightarrow \text{divides}(x, \text{square}(y))$
- No existential quantifier

Conversion to horn sentences

- Existential quantifiers can be removed using existential elimination (Skolemization)
- If the existential quantifier is outside any universal quantifier, a skolem constant is introduced e.g. $\exists y \text{ prime}(p) \Rightarrow \text{prime}(p)$, where p is a Skolem constant
- Otherwise a skolem function is introduced e.g.
 $\forall x \exists y \text{ prime}(y) \wedge \text{divides}(x, y) \Rightarrow \forall x \text{ prime}(\text{PD}(x)) \wedge \text{divides}(x, \text{PD}(x))$, where $\text{PD}(x)$ is a Skolem function

Convert to HS

- And elimination
 - $\text{Prime}(p) \wedge \text{divide}(x,p)$ can be written as two clauses
 - Prime (p)
 - Divides (x,p)
 - Any atomic sentence is a horn clause

Substitution

- Substitution replaces variables with constant
 - $\text{SUBST}((x/49), \text{perfect_sq}(x)) = \text{Perfect_sq}(49)$
 - $\text{SUBST}((x/49, y/7), \text{Divides}(x,y)) = \text{Divides}(49, 7)$

unification

- Unification is the process of finding a substitution that makes two atomic sentences identical (replace with another)
 - UNIFY(Prime(7), prime (x) = (x/7)
 - UNIFY (Divides(49,x) divides (y, 7) = (y/49, x/7)
 - UNIFY (Prime(6) , prime (16) impossible
 - UNIFY (Divides(49,x) divides (x, 7) impossible

Resolution in FOL

- Resolution is an inferencing method that is sound
- Revisit

A few statements

- All people who are graduating are happy
- All happy people smile
- Someone is graduating
- Is someone smiling?

Solving the problem

- We intend to code the problem in predicate calculus
- Use resolution refutation to solve problem
- Solving = whether the conclusion can be answered from the given set of sentences [KB]

Selecting the predicates

Encoding sentences in predicate logic

Predicates

Converting to clausal form

Converting to canonical form

- Step 2: reduce the scope of negation

Converting to canonical form

- Step 3: standardize variables apart (every clause has distinct variable)

Converting to canonical form

- Step 4: move all quantifiers to the left

Converting to canonical form

- Step5: eliminate existential quantifier

Converting to canonical form

- Step 6: Drop all $\forall$

Converting to canonical form

- Step 7: convert to conjunct of disjunct form
- Step 8: make each conjunct a separate clause
- Step 9: standardize variables apart again

Resolution

- What is the major difference in resolution in PL and resolution in FOL?

